Daniel Zhang

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Education

University of Waterloo *Waterloo, ON*
PhD in Statistics; Supervisor: Prof. Ali Ghodsi; graduate average: 91% *Jan. 2025 – expected 2028*

- Research areas: Bayesian deep learning, variational inference, machine learning theory, uncertainty quantification, and efficient language models.
- Completed the first doctoral comprehensive examination through projects on Bayesian backpropagation-free neural networks and uncertainty-aware aggregation of stochastic LLM generations.

University of Waterloo *Waterloo, ON*
Master of Mathematics in Statistics; Supervisor: Prof. Ali Ghodsi; graduate average: 91% *Sept. 2023 – Aug. 2024*

- Research project: *GAM SymbolicGPT: A Generalized Additive Model Approach to Symbolic Regression*.

University of Waterloo *Waterloo, ON*
Honours Bachelor of Mathematics in Statistics; Minor in Computing; Co-operative Program *Sept. 2018 – Apr. 2023*

Research Experience and Contributions

University of Waterloo *Waterloo, ON*
Doctoral Researcher, Data Analytics Laboratory; Supervisor: Prof. Ali Ghodsi *2025 – present*

- Research in Bayesian deep learning, variational inference, uncertainty quantification, and efficient language models.

University of Waterloo *Waterloo, ON*
Doctoral Comprehensive Research Projects; Supervisor: Prof. Ali Ghodsi *2025*

- **Bayesian NoProp:** introduced Bayesian linear and convolutional layers into a backpropagation-free learning framework, enabling decomposition of epistemic and aleatoric uncertainty through variational inference and Monte Carlo prediction.
- Studied KL annealing, label embeddings, calibration, and distribution shift on CIFAR-10/100; found that excessive KL regularization can induce posterior collapse and near-random performance in wider or harder settings.
- **Trust-Weighted LLM Aggregation:** formulated stochastic decoding as sampling over implicit reasoning paths and used accuracy- and embedding-variance-based trust scores to aggregate generations and assess entropy, calibration, and semantic disagreement.

University of Waterloo *Waterloo, ON*
Graduate Researcher, Data Analytics Laboratory; Supervisor: Prof. Ali Ghodsi *Sept. 2023 – Aug. 2024*

- Proposed *GAM SymbolicGPT*, extending neural symbolic regression with additive structure and a backfitting-inspired procedure to reduce the search burden in higher-dimensional problems.
- Designed experiments on synthetic and applied regression tasks to characterize where additive decomposition improves interpretability and where interactions limit the approach.

Vision and Image Processing Laboratory, University of Waterloo *Waterloo, ON*
Undergraduate Researcher; Supervisors: Prof. Alexander Wong, Dr. Jason Deglint, and Vikram Voleti *Sept. – Dec. 2022*

- Led *Plankton-FL*, a privacy-preserving federated learning study for phytoplankton classification across heterogeneous data sources; served as first author and presented the work at CVIS 2022.
- Investigated optimization, client participation, and model-alignment strategies, including Git Re-Basin, for learning across disjoint image domains.

National Research Council Canada *Toronto, ON*
Research Assistant; Supervisors: Dr. Mohammad Sajjad Ghaemi and Dr. James H. K. Ooi *Jan. – Apr. 2021*

- Developed a two-stage transfer-learning framework that pre-trained on influenza records and fine-tuned on COVID-19 clinical data for patient-severity prediction.
- Improved AUROC by 25% relative to an MLP trained from scratch and addressed class imbalance through clinical feature engineering and SMOTE.

Publications

Peer-reviewed publication

Zhang, D., Voleti, V., Wong, A., and Deglint, J. (2022). "Plankton-FL: Exploration of Federated Learning for Privacy-Preserving Training of Deep Neural Networks for Phytoplankton Classification." *Proceedings of the 8th Annual Conference on Computational Vision and Intelligent Systems*. **Best AI Paper Award**.

Awards, Fellowships, and Distinctions

2026	Statistics and Actuarial Science Chair's Graduate Award, University of Waterloo
2024–2026	Graduate Research Studentship, University of Waterloo
2022	Best AI Paper Award, Waterloo AI Institute
2018	President's Scholarship, University of Waterloo

Presentations and Selected Research Training

Zhang, D. (2026). "Tune Small, Train Big: BNN-μP for Bayesian Neural Network Language Models."
Oral presentation, Waterloo Student Conference in Statistics, Actuarial Science and Finance,
University of Waterloo, Waterloo, ON, October 23, 2026.

Zhang, D., Voleti, V., Wong, A., and Deglint, J. (2022). "Plankton-FL: Exploration of Federated Learning for Privacy-Preserving Training of Deep Neural Networks for Phytoplankton Classification." Oral presentation, Computational Vision and Intelligent Systems Conference, Waterloo, ON.

Selected participant, *Princeton Machine Learning Theory Summer School*, Princeton University, Aug. 2026.

Teaching and Academic Leadership

University of Waterloo *Waterloo, ON*
Head Teaching Assistant *2024, 2026*

- STAT 940 (Deep Learning) and STAT 442/842 (Data Visualization); supported course delivery, assessment coordination, and consistent feedback across undergraduate and graduate cohorts.

University of Waterloo *Waterloo, ON*
Teaching Assistant *2023 – 2026*

- Supported ten course offerings spanning statistical learning, advanced regression, case studies in data science, computational statistics, forecasting, data visualization, and introductory statistics.
- Developed grading guidance and detailed feedback for technical reports, case studies, presentations, and statistical analyses.

Industry and Applied Data Science Experience

Intact Financial Corporation *Toronto, ON*
Data Scientist I, Usage-Based Insurance *June 2023 – Dec. 2024*

- Analysed more than 500,000 telematics trips and over 100 candidate features; reduced the active feature set by 25% while improving the driving-risk model's Gini coefficient by 2%.
- Collaborated with data scientists, developers, and product teams on production driving-event logic and validation for mobile features used by more than 100,000 users.

Intact Financial Corporation *Toronto, ON*
Data Science Intern, Usage-Based Insurance *Sept. – Dec. 2022; Sept. – Dec. 2021*

- Built PySpark and Pandas pipelines for commercial-vehicle and insurance-pricing applications, integrating high-volume telematics data with claims and actuarial modelling workflows.
- Performed statistical feature screening, refactored production pipelines, and created modular preprocessing tools for custom actuarial features.

TD Bank *Toronto, ON*
Data Science and Advanced Analytics Intern *May – Aug. 2022*

- Automated legacy analytics workflows and developed SQL/PySpark data pipelines for TD Wealth; contributed to migration of ETL processes to Azure-based infrastructure.

Technical Skills

Research	Bayesian neural networks, variational inference, uncertainty quantification, μP and scaling laws, Transformers, calibration, distribution shift
Programming	Python, PyTorch, NumPy, Pandas, scikit-learn, TensorFlow/Keras, R, SQL, PySpark, SAS
Computing	SLURM, multi-GPU and distributed training, Linux, Git, high-performance computing, LaTeX